THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2012 Fundamental Concepts of Mathematics
Final Exam

Answer ALL Questions

- You need to scan your work and submit it in Moodle as a PDF file before Dec 18 (Wed), 9:30 a.m. (Hong Kong Time). **No late work will be accepted**.
- You can only submit your work ONCE before the closing time.
- Make sure that your work is properly scanned. Oversized or blurred images will NOT be graded.
- You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write**.

This is an open book test. You may consult the lecture notes and other materials from Moodle. However, no collaboration is permitted.

1. (*30 points*)

Prove or disprove each of the following statements:

(a) There exists a prime $p \geq 5$ such that $p^2 \not\equiv 1 \pmod{24}$.

(b) There exist integers x and y such that $x^4 + y^4 = 2019$.

(c) For any $m, n \in \mathbb{N}$ with $m > n$, if $\sqrt{m} + \sqrt{n}$ or $\sqrt{m} - \sqrt{n}$ is rational, then both m and n are perfect squares.

(d) Let A, B be nonempty sets, U, V be subsets of $A \times B$, and

$$X = \{a \in A : (a, b) \in U \text{ for some } b \in B\}, \quad Y = \{a \in A : (a, b) \in V \text{ for some } b \in B\}.$$

If $U \cup V = A \times B$, then $A = X \cup Y$.

(e) Let A be a nonempty set, and let P_1 and P_2 be two partitions of A. Define

$$P = \{S \in \mathcal{P}(A) \setminus \{\emptyset\} : S = S_1 \cap S_2 \text{ for some } S_1 \in P_1 \text{ and } S_2 \in P_2\}.$$

Then P is a partition of A.

2. (*14 points*)

For any $n \in \mathbb{Z} \setminus \{0\}$, define

$$n\mathbb{Z} = \{x \in \mathbb{Z} : x = nk \text{ for some } k \in \mathbb{Z}\}.$$

(a) Prove that $n\mathbb{Z} = (-n)\mathbb{Z}$ for any $n \in \mathbb{Z} \setminus \{0\}$,

(b) For any $n, m \in \mathbb{Z} \setminus \{0\}$, give a necessary and sufficient condition for

$$n\mathbb{Z} = m\mathbb{Z}.$$

Provide a detailed proof to justify your claim.

3. (*20 points*)

Consider the sequence $F_1, F_2, F_3, \ldots$ of *Fibonacci numbers* satisfying $F_1 = 1$, $F_2 = 1$ and

$$F_n = F_{n-1} + F_{n-2} \quad \text{for any integer } n \geq 3.$$

Use induction to prove that

(a) for any positive integers m and n,

$$F_{m+n+1} = F_{n+1}F_{m+1} + F_nF_m.$$

(b) every positive integer n can be expressed as a sum of one or more distinct terms in the Fibonacci sequence.

4. (*24 points*)

Let A and B be nonempty sets and let $f : A \longrightarrow B$ be a function. Consider the following statements:

(1) $B \setminus f(S) \subseteq f(A \setminus S)$ for any subset S of A.

(2) $f(A \setminus S) \subseteq B \setminus f(S)$ for any subset S of A.

(3) For any subsets S and T of A, if $f(S) = f(T)$, then $S = T$.

(4) For any subsets U and V of B, if $f^{-1}(U) = f^{-1}(V)$, then $U = V$.

Exactly two of the above statements are necessary and sufficient conditions for "f is injective" and the other two are necessary and sufficient conditions for "f is surjective". Make all the four assertions and prove each of them.

5. (*12 points*)

Let A be a nonempty finite set and let $f : A \longrightarrow A$ be a function. Suppose that $f \circ f = id_A$, the identity function on A.

(a) Let R be a relation on A defined by

$$aRb \text{ if } a = b \text{ or } f(a) = b.$$

Prove that R is an equivalence relation on A. Find also the equivalence class $[a]$ for any $a \in A$.

(b) Denote by A_f the set of *fixed points* of f, that is,

$$A_f = \{a \in A : f(a) = a\}.$$

Use (a), or otherwise, prove that $|A|$ and $|A_f|$ have the same parity.

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